perplexity

i don't think its straight a copy clone from my traits we don't need like a separtae stitch to design right ..? So draft the v0.4.0 prd then ..

Correct—no separate Stitch redesign is needed. Extend the existing Trails Journal and Map components, keeping the same visual language and interactions. Discover introduces a different source of trails and the controls needed to explore them—not a second app.

Trails — Product Requirements Document

v0.4.0 · Discover, Save, and Follow Shared Trails

Status: Implementation specification
Platform: Native iOS
Backend dependency: Separate `trails.logos` project
Foundation: Existing capture, journal, map, playback, and validated retrace capabilities

> Discover a place through someone else's experience. Keep it, revisit it, and add your own perspective.

1. Release objective

Trails v0.4.0 extends the working personal application into a community experience.

Users can:

- Discover published trails geographically.
- Find trails by creator username.
- Combine area and creator filters.
- Browse the same results in Journal or Map.
- Bookmark trails without downloading them.
- Download and import complete trails.
- Revisit their actual media.
- Retrace when the available positioning data and alignment checks support it.
- Publish their own saved trails through `trails.logos`.

The central design principle is **continuity**: reuse the application that already works rather than build a separate discovery product beside it.

2. Product boundaries

Native Trails app owns

- Existing capture experience.
- Personal recordings and downloaded library.
- Discover screens.
- Filters and bookmarks.
- Download/import presentation.
- Publication preparation and progress.
- Media playback.
- Map and spatial viewing.
- Live alignment and retrace.
- Recording a new perspective.

trails.logos owns

- Package validation.
- Logos Storage upload and retrieval.
- Authoritative publication registry on LEZ.
- Public catalogue reads and discovery queries.
- Publication revisions and listing status.
- Confirmed publication receipts.

The app must not invent publication state when the backend is unavailable.

Out of scope

- Followers, comments, ratings, and popularity scores.
- General account onboarding.
- Public identity verification.
- User-wallet onboarding.
- AI-generated content.
- Turn-by-turn navigation.
- Automatic alignment between independent captures.
- Automatic merging of perspectives.
- A new visual design system.

3. Navigation and component reuse

Main destinations

Destination	Contents
My Trails	Personal recordings, downloaded trails, and recoverable drafts
Discover	Published catalogue, geographic discovery, creator discovery, and bookmarks

Keep the existing Start a trail action prominent.

Settings remains in the existing location.

Shared presentation

Both destinations reuse:

- Journal layout.
- Map renderer.
- Trail cards.
- Moment indicators.
- Cover treatment.
- Selected-trail sheet.
- Trail detail.
- Playback.
- Existing spatial views.

The context changes the data and available actions.

```
My Trails
  Source: records on this device
  Views: Journal / Map

Discover
  Source: public catalogue
  Views: Journal / Map
  Filters: area / creator / saved
```

Do not duplicate viewer implementations.

Design process

Use current app screens as the implementation reference.

Stitch is optional for resolving a specific layout gap. It is not a prerequisite and must not trigger a redesign of working screens.

4. Core user journeys

Discover geographically

```
Open Discover
→ Choose Map
→ Select an area
→ Search this area
→ Select a trail
→ Read detail
→ Bookmark or download
```

Discover by creator

```
Open Discover
→ Search/select username
→ View that creator’s trails
→ Optionally constrain to an area
→ Open a result
```

Keep and revisit

```
Bookmark
→ Return through Saved
→ Download
→ Import
→ Open in existing viewer
→ Revisit or attempt retrace
```

Publish

```
Open an owned saved trail
→ Review publication
→ Confirm included data
→ Upload and register
→ Verify publication
→ Open public detail
```

5. Discover screen

Shared header

```
Discover

[Search trails, places, or creators]

[Area] [Creator] [Saved]

[Journal | Map]
```

The exact arrangement should follow existing component patterns.

Requirements

- Area and creator controls have equal prominence.
- Active filters remain visible.
- Journal and Map share one query state.
- Switching views preserves filters and selection.
- Returning from detail preserves browsing context.
- Results use actual publication records.
- Do not add “Trending” or “Popular” without defined supporting data.

Initial state

Restore the last valid discovery context where possible.

For a first visit:

- Offer Near me.
- Allow selecting an area without location permission.
- Allow browsing public trails in Journal without an area restriction.

Do not silently fabricate a location.

6. Area filtering

Supported choices

Mode	Behaviour
All areas	No geographic restriction
Near me	Query around an available permitted location
This map area	Query the selected viewport
Chosen place	Move to a selected place and query that area

Map interaction

- Moving the map does not continually replace results.
- Show Search this area after meaningful movement.
- Tapping it updates the geographic query.
- Preserve other filters.
- Show the selected search area consistently in Journal.

Geographic inclusion rule

For v0.4.0:

> A trail matches an area when its published discovery anchor falls within that area.

This is not a claim that the entire recorded route lies inside the area.

Trails without an anchor

They:

- Remain available in unrestricted Journal results.
- Remain available through creator queries.
- Can be bookmarked and downloaded.
- Do not appear at invented map coordinates.
- Do not match area-constrained results.

When relevant, explain that some results have no map location.

7. Creator and username filtering

Entry points

- Creator filter.
- Search results.
- Creator attribution on a trail card or detail.

Required identity fields

```
creatorId
username
displayName
publisherReference
```

Filter using `creatorId`, not mutable display-name text.

Combined filtering

```
Area: Selected map region
Creator: @mira
Saved: On
```

Results must match all active filters.

Provide:

> Show all trails by @mira

This removes the area restriction while retaining the creator.

Identity limitations

- Usernames are attribution unless uniqueness is enforced.
- Do not show verified badges.
- Creator selection can be a sheet; a separate profile screen is not required.
- The app must handle more than one creator record, even if initial publishing uses one funded account.

Keep the signing publisher reference and creator attribution distinct. A shared publishing service must not make every trail appear to have the same human creator.

8. Search behaviour

The shared search entry distinguishes:

- Trails.
- Creators.
- Places.

Selecting:

- A trail opens detail.
- A creator applies its filter.
- A place sets the geographic context.

Existing filters remain unless the chosen action explicitly changes them.

Implementation contract

- Trail and creator search use `trails.logos`.
- Place search uses the configured map/geocoding integration.
- Do not send creator queries to the place-search provider.
- Do not expose live suggestions unless the chosen service supports that flow.

- If only submitted search is implemented, make that interaction clear.

9. Journal presentation

Reuse the existing journal cards.

Public card fields

- Actual title.
- Actual cover or neutral placeholder.
- Creator attribution.
- Optional place label.
- Recording date.
- Content indicators.
- Bookmark control.
- Downloaded state when applicable.

Sorting

Required:

- Recently published.

Optional:

- Nearest, when a reference location and supported query are available.

Performance

- Do not load full trail packages for cards.
- Use separately retrievable previews when available.
- Use neutral placeholders otherwise.
- Paginate results.

No fabricated thumbnails or media counts.

10. Map presentation

Reuse the existing map and selected-trail sheet.

Required behaviour

- Show geographically located results.
- Cluster or group overlapping markers.
- Select a marker to open its result card.
- Open full detail from the card.
- Recenter when location is available.
- Fit current results.
- Preserve selection when switching to Journal.

Multiple trails at one location

Show a group or list of independent trails.

Do not:

- Overwrite one perspective with another.
- Merge geometry.
- Infer shared spatial frames from proximity.

The map provider supplies geographic context. The Trails catalogue supplies trail markers and metadata.

11. Bookmarks

Bookmarking is independent of downloading.

Action	Outcome
Bookmark	Save a reference for later
Download	Store the package locally
Remove bookmark	Remove the saved reference
Remove download	Remove local package content

Requirements

- Bookmark from cards and detail.
- Store locally for this release.
- Restore after application restart.
- Saved filter works in Journal and Map.
- Removing a bookmark does not delete downloaded content.
- Removing a download does not remove its bookmark.

Suggested feedback:

> “Bookmarked. Download to view offline.”

Unavailable publications

Keep the bookmark with an unavailable state rather than silently removing it.

Do not imply bookmarks synchronise across devices.

12. Public trail detail

Reuse the existing detail structure, adapted for a remote trail.

Show before download

- Title.
- Creator.
- Introduction.
- Place, if supplied.
- Recording date.
- Contents and capability summary.
- Package size.
- Bookmark state.
- Download state.

Do not imply media previews are playable before download unless remote playback is implemented.

Primary action

State	Action
Not downloaded	Download trail
Downloading	View progress
Checking/importing	Preparing trail
Downloaded	Open trail
Failed	Try again
New revision detected	Update available

Secondary publication details

- Trail ID.
- Revision.
- Package CID.
- Registry reference.
- Publisher reference.
- Network.

Keep these out of the primary browsing hierarchy.

13. Download and import

Lifecycle

```
Queued
→ Downloading
→ Validating
→ Importing
→ Ready
```

Additional states:

- Cancelled.
- Failed.
- Update available.

Required behaviour

- Show actual byte progress when available.
- Use stage labels instead of invented percentages.
- Validate package schema and asset references.
- Import into a separate managed directory.
- Commit the local library record only when import is usable.
- Preserve published provenance.
- Keep temporary data separate from completed trails.

Completion

> “Trail ready on this iPhone.”

Actions:

- Open trail.

- Back to Discover.

Failure handling

Explain:

- Network failure.
- Package unavailable.
- Insufficient space.
- Unsupported package version.
- Invalid or incomplete package.

Do not create a normal-looking library item backed by a failed import.

Pause/resume is not required. Do not label Cancel as Pause.

14. Downloaded trails and provenance

The existing viewer must open downloaded content.

Preserve:

```
sourceTrailId
sourceRevision
sourcePackageCid
creatorId
publisherReference
downloadedAt
```

Ownership rules

- The downloaded source is not an owned recording.
- Local actions do not modify the public publication.
- Retracing creates a separate session.
- Recording a perspective creates a new personal trail.
- Source attribution remains visible.

My Trails filters

```
All | Recorded by me | Downloaded
```

Downloaded content must remain accessible without Discover being online.

15. Revisit and retrace

Revisit

Use existing playback for:

- Photos.
- Videos.
- Voice.
- Notes.
- Available maps, plans, and geometry.

Retrace

Retain all v0.3.1 requirements:

- Establish the reference frame.
- Verify alignment.
- Display uncertainty and tracking loss.
- Keep the original unchanged.
- Offer manual follow-along.

Downloading alignment assets does not establish live alignment.

Capability language

Prefer:

- Recorded outdoor path.
- Indoor plan included.
- Indoor alignment data included.
- Manual follow-along available.

Avoid guaranteed navigation claims.

16. Record your perspective

Provide a secondary action:

> Record your perspective

Behaviour

- Start a new personal trail.
- Preserve an optional source-publication reference.
- Reuse a place label only when appropriate.
- Capture new media independently.
- Do not copy source media.
- Do not assume coordinate alignment.

Public perspective-grouping is not required for this release. Store the relationship locally and include it in exports when the package schema supports it.

17. Publishing owned trails

Entry

Owned, saved trail → Publish.

Downloaded trails cannot be republished as the user's original recording.

Review

Show:

- Title.
- Introduction.
- Creator username/display name.
- Cover.
- Included media.
- Included geographic/spatial data.
- Public-data acknowledgement.

Progress

```
Preparing
→ Uploading
→ Verifying storage
→ Registering on LEZ
→ Confirming
→ Published
```

Success condition

The backend confirms that the authoritative publication record points to the verified package.

Return:

- Trail ID.
- Revision.
- Package CID.
- Registry/transaction references.

Provide View public trail.

Do not show Share link unless a working destination or deep-link flow exists.

18. Required backend contract

`trails.logos` must expose enough metadata to browse without downloading archives.

Discovery record

```
DiscoveryRecord
  trailId
  revision
  title
  shortDescription?
  creatorId
  username
  displayName
  publisherReference
  recordedAt
  publishedAt
  placeLabel?
  discoveryAnchor?
  coverReference?
  capabilities
  packageCid
  packageSizeBytes
  listed
```

Queries

Proposed application interfaces:

```
listTrails(
  area?,
  creatorId?,
  query?,
  cursor?,
  limit?
```

```
)

searchCreators(query)
getTrail(trailId)
getPublicationStatus(operationId)
downloadTrail(trailId, revision)
```

Authority

- LEZ remains authoritative for publication state.
- Discovery indexes are derived and rebuildable.
- Bookmarks are local.
- Imported package state is local.
- Creator attribution has an explicit relationship to the publisher.

19. State preservation

Maintain a shared discovery state:

```
DiscoveryState
  presentationMode
  areaFilter
  creatorFilter
  bookmarkedOnly
  query
  sort
  selectedTrailId
  mapViewport
  journalScrollPosition
```

Requirements:

- Preserve state when opening and closing detail.
- Preserve filters when switching Journal/Map.
- Do not reset the map after bookmarking.
- Avoid duplicate download operations.
- Restore sensible context after returning from playback.

20. Empty, offline, and error states

Situation	Required response
No public records	Honest empty catalogue
No filter matches	Explain active filters; offer individual removal

Situation	Required response
Location denied	Allow manual area selection
Offline	Keep local recordings and downloads available
Registry unavailable	Retry; do not fabricate results
Missing cover	Neutral placeholder
Missing map location	Journal availability without fake marker
Unlisted bookmarked trail	Unavailable/listing status
Unsupported alignment data	Playback plus manual follow-along

Never silently broaden a filtered query.

21. Implementation priorities

P0 — Connected discovery

- Reuse Journal and Map.
- Area filtering.
- Creator filtering.
- Combined filter state.
- Public detail.

P0 — Usable shared content

- Download.
- Validation/import.
- Open in existing viewer.
- Provenance.
- Offline reopening.

P0 — Publishing

- Publish review.
- Real progress.
- Confirmed public result.

P1 — Return and participation

- Bookmarks.
- Saved filter.
- Record your perspective.
- Update-available state.

For the release to meet the agreed scope, bookmarks must be complete before sign-off; the priority order only determines implementation sequence.

Deferred polish

- Public creator pages.
- Rich place pages.
- Advanced ranking.
- Streaming previews.
- Complex revision management.

22. Acceptance scenarios

A. Area discovery

Select an area, search it, and open a real result.

Expected: results match the documented discovery-anchor rule.

B. Creator discovery

Select a username and browse its trails.

Expected: filtering uses stable creator identity.

C. Combined filters

Apply area, creator, and Saved.

Expected: all conditions apply in both Journal and Map.

D. Context continuity

Open detail, return, switch presentation.

Expected: filters and browsing position remain coherent.

E. Bookmark lifecycle

Bookmark, restart, find in Saved, remove bookmark.

Expected: no automatic download or content deletion.

F. Clean-install consumption

Discover, download, import, play media, restart offline.

Expected: actual content remains usable.

G. Retrace honesty

Open a downloaded indoor trail.

Expected: live positioning appears only after validated alignment.

H. Publication

Publish an owned trail and find it from another installation.

Expected: verified backend state, not a simulated success.

I. Ownership

Open a downloaded trail and record a perspective.

Expected: a new independent recording with source attribution, not overwritten content.

23. Definition of done

Trails v0.4.0 is ready when someone can:

> Find a trail by place or person, save it, download it, experience it through the existing viewer, and contribute an independent perspective.

It should feel like the existing Trails app opening outward—not a separate app bolted onto it.